

Supplementary Material for: An open-source GCR dosimetry pipeline for Mars mission risk assessment: organ-specific cancer risk, acute solar particle events, and global uncertainty quantification

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ABSTRACT

This document contains full physics/biology derivations, extended parameter tables, and extended discussion supporting the main text, which is constrained to 3500 words (Introduction through Discussion) per the *Life Sciences in Space Research* author guidelines. Section, table, and equation numbers here are independent of the main text and prefixed with S.

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1. Supplementary Methods

1.1. Mission Trajectory Model

A Hohmann-transfer trajectory from Earth (1.0 AU) to Mars (1.524 AU) is generated by computing heliocentric distance as a function of mission elapsed time, $r(t)$, using Keplerian orbital mechanics:

$$r(t) = \frac{a(1 - e^2)}{1 + e \cos \theta(t)}, \quad (1)$$

where $a = (r_{\oplus} + r_{\text{Mars}})/2 = 1.262$ AU is the semi-major axis and $e = (r_{\text{Mars}} - r_{\oplus})/(r_{\text{Mars}} + r_{\oplus}) = 0.208$ is the eccentricity of the transfer ellipse. Transit duration is fixed at 259 days (one half-period of the transfer orbit). The heliocentric distance enters the dose rate only through the solar modulation potential $\phi(t)$; direct geometric flux dilution as r^{-2} is not applied separately because the force-field modulation already encodes the energy dependence of solar wind effects at the relevant energies.

1.2. Solar Modulation Potential

The time-resolved solar modulation potential $\phi(t)$ is obtained from the Usoskin et al. (2017) database, which provides monthly ϕ values reconstructed from ground-level neutron monitor count rates back to 1936. For the MSL transit period (2011 November – 2012 July), values cluster around $\phi \approx 481 \pm 50$ MV. Systematic offsets between different neutron monitor stations introduce ± 50 –100 MV uncertainty in the absolute ϕ calibration; this is propagated via the `phi_scale` Latin Hypercube Sampling (LHS) parameter (Table 6).

1.3. GCR Spectrum Generation

1.3.1. Force-Field Modulation

Solar modulation of the galactic cosmic ray (GCR) flux is computed using the Gleeson–Axford (1968) force-field approximation. For an ion of charge Z and mass number A , the modulated differential flux at kinetic energy per nucleon E is:

$$J(E, \phi) = J_{\text{LIS}}(E + \delta E) \cdot \frac{E(E + 2E_0)}{(E + \delta E)(E + \delta E + 2E_0)}, \quad (2)$$

where $E_0 = 938.272$ MeV is the proton rest-mass energy, $\delta E = (Z/A)\phi$ (MeV/nucleon), and J_{LIS} is the local interstellar spectrum (LIS, below). The ratio in Eq. (2) is the Jacobian of the energy transformation, ensuring flux conservation. The approximation is accurate for rigidities above ~ 1 GV and ϕ in 200–1500 MV (Potgieter, 2013); charge-sign-dependent drift (relevant at solar minimum) is not captured, but dose here is dominated by particles above a few hundred MeV/n where this is not expected to matter materially.

1.3.2. Local Interstellar Spectra

Proton and helium LIS follow Vos & Potgieter Vos and Potgieter (2015):

$$J_{\text{LIS}}^{\text{H}}(E) = C_{\text{H}} \cdot \frac{(E_{\text{GeV}} + 0.67)^{-3.93}}{\beta^2}, \quad (3)$$

with $C_{\text{H}} = 1.9 \times 10^4 \text{ cm}^{-2} \text{ s}^{-1} \text{ MeV}^{-1} \text{ sr}^{-1}$. Heavy ions (C, O, Si, Fe) follow the per-species HelMod model of Boschini et al. Boschini et al. (2020):

$$J_{\text{LIS}}^{(s)}(E_n) = C_s \cdot \frac{(E_{\text{GeV/n}} + \delta_s)^{-\alpha_s}}{\beta^2}. \quad (4)$$

Table 1 lists all adopted parameters. Iron’s flatter index ($\alpha_{\text{Fe}} = 2.65$) gives it disproportionate high-energy flux, amplified by Z^2 stopping-power scaling.

1.3.3. Composition-Constrained Normalization

Calibration is a two-step procedure. **Step 1 (composition constraint)**: for each species s , a correction factor $r_s = (J_s/J_{\text{H}})_{\text{ACE/CRIS}}/(J_s/J_{\text{H}})_{\text{LIS}}$ is computed so the modulated LIS reproduces the particle-flux ratio measured by ACE/CRIS (George et al., 2009) and PAMELA (Adriani et al., 2014) at $E_n = 200$ MeV/n, $\phi = 481$ MV (Table 2).

Table 1

LIS spectral parameters used in `gcrisk/spectrum.py`. Proton and helium from Vos & Potgieter (2015); heavy ions from Boschini et al. (2020).

Species	C_s	α_s	δ_s (GeV/n)	Primary source
H	1.90×10^4	3.93	0.67	Vos & Potgieter (2015)
He	1.82×10^3	2.78	0.62	Vos & Potgieter (2015)
C	2.50×10^2	2.72	0.50	Boschini et al. (2020)
O	1.80×10^2	2.69	0.50	Boschini et al. (2020)
Si	3.00×10^1	2.70	0.50	Boschini et al. (2020)
Fe	4.30×10^0	2.65	0.50	Boschini et al. (2020)

Table 2

ACE/CRIS and PAMELA reference particle-flux ratios relative to $J_H = 1.0$, $E_n \approx 200$ MeV/n, $\phi \approx 481$ MV (2011–2012 MSL epoch).

Species	J_s/J_H	Source
He	6.60×10^{-2}	Adriani et al. (2014)
C	1.80×10^{-3}	George et al. (2009)
O	2.20×10^{-3}	George et al. (2009)
Si	1.80×10^{-4}	George et al. (2009)
Fe	1.40×10^{-4}	George et al. (2009)

Table 3

Per-charge-group absorbed dose fractions: composition-constrained pipeline vs. NSRL reference (Slaba et al., 2016). Geometries differ by 4 g cm^{-2} Al and solar phase; evaluating the pipeline at NSRL conditions (20 g cm^{-2} Al, $\phi \approx 300$ MV) gives HZE = 12.6% vs. NSRL's 7.6%.

Particle class	Pipeline (16 g cm^{-2} Al, $\phi = 481$)	NSRL (20 g cm^{-2} Al, solar min)
$Z = 1$ (protons)	68.9%	73.3%
$Z = 2$ (helium)	9.8%	19.2%
$Z > 2$ (HZE)	13.0%	7.6%

Step 2 (global dose scale): a single global scale $k = 0.894$ is then fit so $J_s \leftarrow k \cdot r_s \cdot J_s$ reproduces $\dot{D} = 1.84 \text{ mGy day}^{-1}$ at 16 g cm^{-2} Al, $\phi = 481$ MV (Zeitlin et al., 2013). This replaces a prior six-free-parameter fit (one factor per species fit independently to the single dose scalar, with no physical composition constraint) with one free parameter whose relative species mix is externally anchored.

Table 3 compares the resulting composition against the NASA Space Radiation Laboratory (NSRL) GCR simulator reference field (Slaba, Blattnig, Norbury, Rusek and La Tessa, 2016). The proton fraction is within ~ 4 points of NSRL; the high-charge, high-energy (HZE) fraction is $\sim 1.7\times$ the NSRL reference (down from $\sim 4\times$ under the prior calibration), attributed to continuous-slowing-down approximation (CSDA) transport not modeling nuclear fragmentation (which converts high linear-energy-transfer (LET) heavy ions into lower-LET secondaries in NSRL/HZETRN but not here). The helium fraction is below NSRL, reflecting the measured ACE/CRIS He/H ratio at 200 MeV/n being lower than the LIS model's He/H at the higher energies that set the unmodulated normalization.

With this composition, the pipeline gives $\dot{H} = 6.09 \text{ mSv day}^{-1}$ and $Q_{\text{eff}} = 3.32$ at the calibration geometry, vs. RAD's $4.81 \pm 0.97 \text{ mSv day}^{-1}$ and 2.62 ± 0.14 (a 27% excess). This is attributed to the same CSDA fragmentation gap: without converting high-LET HZE dose into lower-LET secondaries, the dose-weighted quality factor is inflated. $\dot{H}$ and organ-specific cancer risk (REID) from this pipeline should be read as conservative upper estimates. The `hze_norm_scale` LHS parameter (LogNormal, $\sigma = 0.20$) propagates the residual $\sim 20\%$ ACE/CRIS inter-instrument composition uncertainty (George et al., 2009; Boschini et al., 2020).

Table 4

Organ tissue depths and ICRP-60 tissue weighting factors. Depths from ICRP Publication 89 (ICRP, 2002).

Organ	Depth (g/cm ²)	w_T (ICRP-60)	Cancer type modeled
Stomach	8.0	0.12	Gastric carcinoma
Colon	9.5	0.12	Colorectal carcinoma
Lung	5.0	0.12	Lung carcinoma
Leukemia	4.0	0.12	Myeloid leukemia
Esophagus	4.0	0.05	Esophageal carcinoma
Liver	6.0	0.04	Hepatocellular carcinoma
Bladder	10.0	0.04	Transitional cell carcinoma
Thyroid	2.0	0.04	Thyroid carcinoma
Breast	1.5	0.05	Breast adenocarcinoma
Ovary	12.0	0.20	Ovarian carcinoma
Other	7.0	0.05	Composite remainder

1.4. Charged Particle Transport

1.4.1. Stopping Power and CSDA Range

Energy loss is computed via CSDA. Stopping power for charge Z , velocity β :

$$S(E/A, Z) = z_{\text{eff}}^2(E/A, Z) \cdot S_p(E/A), \quad (5)$$

with Barkas effective charge (Barkas, 1963) $z_{\text{eff}} = Z[1 - \exp(-125\beta Z^{-2/3})]$ (reducing to Z^2 scaling above ~ 1 GeV/n). Proton stopping powers (Al, PE, water, tissue) are tabulated from NIST PSTAR/ASTAR (500 log-spaced points, 1 MeV–10⁵ MeV, cubic-spline interpolated). CSDA range $R(E) = \int_0^E dE'/S(E', Z)$; exit energy after slab x is $E_{\text{out}} = R^{-1}(R(E_{\text{in}}) - x_{\text{eff}})$ with $x_{\text{eff}} = x \cdot Z^2/A$; particles with $R(E_{\text{in}}) < x_{\text{eff}}$ stop.

1.4.2. Nuclear Interaction Attenuation

Inelastic nuclear interactions are modeled as exponential fluence attenuation, $\Phi_s(E, x) = \Phi_s^{(0)}(E) \exp(-x/\lambda_s)$, with mean free path λ_s from the Bradt-Peters geometric cross-section (Bradt and Peters, 1950), $\sigma = \pi r_0^2 (A_p^{1/3} + A_t^{1/3} - \delta)^2$ ($r_0 = 1.35$ fm, $\delta = 0.83$). This is the same geometric form used to build HZETRN's tabulated mean free paths (Wilson et al., 1991). This does not model the secondary fragment spectrum; above ~ 30 g cm⁻² for Fe/Si this approximation becomes increasingly crude as fragmentation produces biologically-relevant lighter secondaries (C, O). A full nuclear reaction network (HZETRN, Geant4/TOPAS) would be needed to close this gap. This is the primary physical limitation of this transport model.

1.4.3. Secondary Neutron Production

Rather than tracking individual neutron histories, neutron dose equivalent uses a pre-computed lookup, $\dot{H}_n(x, \phi, \text{mat}) = \dot{H}_n^{\text{Al}}(\phi) \cdot f_{\text{mat}} \cdot g(x)$, with material yield factor f_{mat} (PE $\approx 0.72 \times$ Al) and depth dependence $g(x)$ interpolated from the HZETRN benchmark of Slaba et al. Slaba et al. (2014); neutron $Q_n = 10$ (International Commission on Radiological Protection Publication 60, ICRP-60). This introduces ± 25 –30% uncertainty, propagated via the neutron_H LHS parameter.

1.5. Organ Dose Model

Organ dose is computed by transporting shielded flux through an additional tissue layer at each organ's ICRP Publication 89 depth (ICRP, 2002) (Table 4); total areal density at organ k is $x_{\text{total}}^{(k)} = x_{\text{wall}} + d_k$. Absorbed dose rate:

$$\dot{D}_k = \sum_s \int \Phi_s(E; x_{\text{total}}^{(k)}, \phi) \cdot S_s(E) \cdot \rho^{-1} \cdot C dE, \quad (6)$$

with the 4π isotropic-flux factor absorbed into $C = 1.602 \times 10^{-10}$ Gy per (MeV g⁻¹ cm⁻²); energy grid is 200 log-spaced points, 10–10⁵ MeV/n ($< 0.5\%$ convergence).

Dose equivalent: $\dot{H}_k = \sum_s Q_s \cdot \dot{D}_{k,s}$, with ICRP-60 $Q(L)$ (ICRP, 1991):

$$Q(L) = \begin{cases} 1 & L < 10 \text{ keV } \mu\text{m}^{-1} \\ 0.32 L - 2.2 & 10 \leq L \leq 100 \text{ keV } \mu\text{m}^{-1} \\ 300/\sqrt{L} & L > 100 \text{ keV } \mu\text{m}^{-1} \end{cases} \quad (7)$$

applied per species at the dose-averaged LET, $Q_s = Q(\bar{L}_{D,s})$. This mean-LET approximation (vs. integrating $Q(L)$ over each species' full LET spectrum) underestimates Q for carbon by $\sim 16\%$ and overestimates it for oxygen by $\sim 24\%$ (both straddle the $10 \text{ keV } \mu\text{m}^{-1}$ kink), with $\lesssim 3\%$ bias for H, He, Si, Fe; the opposite-signed C/O biases partially cancel in the field-averaged $Q_{\text{eff}} = \sum H_{k,s} / \sum D_{k,s}$, which is 2.60 at the calibration geometry (before the neutron contribution raises the total apparent Q_{eff} to 3.32), within 0.8% of the RAD ion-only estimate. This is expected, since it follows directly from the calibration procedure.

1.6. Cancer Risk Model (REID) — Full Formulation

REID integral:

$$\text{REID}(a_e, s) = \int_{a_e+L}^{a_{\text{LT}}} \frac{\mu_c(a, s) \cdot S(a | a_e, s)}{S(a_e | s)} da, \quad (8)$$

a_e =age at exposure, $L = 10$ yr minimum latency, $S(a|a_e, s)$ =conditional survival. Excess mortality combines the excess absolute risk (EAR) and excess relative risk (ERR):

$$\mu_c(a) = \lambda_c^{\text{background}}(a) \cdot \text{ERR}(D, a_e) + \text{EAR}(D, a_e, a), \quad (9)$$

with EAR/ERR organ-specific functions of dose, age at exposure, and attained age from Tables 2–3 of Cucinotta et al. Cucinotta, Kim and Chappell (2013). The dose and dose-rate effectiveness factor (DDREF) reduces low-dose-rate risk, $D_{\text{eff}} = D/\text{DDREF}$: point estimates use a fixed DDREF of 1.75; the uncertainty ensemble instead samples $\text{DDREF} \sim \text{Uniform}[1, 2]$. Conditional survival and baseline cancer mortality use interpolated US sex-specific lookup tables in `gcrisk/reid.py`. This is a practical approximation to the NASA/Cucinotta framework.

The REID distribution is right-skewed, driven by `Q_factor`'s heavy-tailed lognormal draws: its $\text{LogNormal}[1.0, \ln 2]$ draws reach a sampled maximum of $\sim 8.2\times$ in the $N = 500$ ensemble, corresponding to $Q_{\text{eff}} \approx 27$ at the calibration point. This approaches but does not exceed the ICRP-60 physical maximum of 30 ($L = 100 \text{ keV } \mu\text{m}^{-1}$). Every LHS sample's `Q_factor` is therefore clipped so effective Q cannot exceed 30; for the reported $N = 500$ ensemble this clip does not bind, so reported percentiles are unchanged, but it removes any possibility of an unphysical excursion at larger N or different seeds.

REID here is computed as a direct hazard-rate integral (Eq. 8). A survival-curve formulation like $1 - \exp(-\mu)$ would confine individual Monte Carlo samples to $[0, 1]$ automatically; this direct-integral form does not. Each sample is therefore also clipped to $[0, 1]$ post hoc, since REID represents a probability and cannot exceed 100% by definition. This second clip barely binds at the reported calibration-geometry ensemble (1 of 500 samples, unclipped maximum 101.3%) and does not move any reported percentile, but it removes draws that are more frequently, and more severely, unphysical at low shielding (REID vs. shielding figure, main text), where the unshielded dose is large enough that the heavy-tailed `Q_factor` draws alone could otherwise push individual samples to 200–300%.

1.7. Solar Energetic Particle Acute Risk

Solar energetic particle (SEP) fluence spectra use the Band function (Tylka and Lee, 2006):

$$J_{\text{SEP}}(E) = C \times \begin{cases} E^{\gamma_\ell} \exp(-E/E_b) & E \leq E_{\text{trans}}, \\ [(\gamma_\ell - \gamma_h)E_b]^{\gamma_\ell - \gamma_h} \exp(\gamma_h - \gamma_\ell) \cdot E^{\gamma_h} & E > E_{\text{trans}}, \end{cases} \quad (10)$$

$E_{\text{trans}} = (\gamma_\ell - \gamma_h)E_b$. Three canonical events (Table 5) use parameters from Tylka & Dietrich Tylka and Lee (2006); Aug 1972 predates space-borne dosimetry and is reconstructed from ground-level neutron monitor/rionetry data, carrying larger uncertainty ($\sim \pm 50\%$) than the 2003/2005 events.

Blood-forming organ (BFO) absorbed dose transports event-integrated proton fluence through wall + 10 g cm^{-2} tissue via CSDA; only protons are tracked (heavy-ion SEP fluence is generally dose-negligible at these energies).

Table 5

Band-function spectral parameters for three canonical SEP events.

Event	C	γ_ℓ	γ_h	E_b (MeV)	Duration (days)
Aug 1972	4.0×10^4	-1.78	-3.98	32	2
Oct 2003	1.5×10^4	-1.40	-3.20	50	3
Jan 2005	3.0×10^2	-0.90	-2.80	110	1

Table 6Latin Hypercube Sampling parameters, distributions, and physical justifications. LogNormal[1.0, σ] means a multiplicative factor $\exp(\mathcal{N}(0, \sigma))$ with median 1.0.

Parameter	Distribution	Justification
phi_scale	LogNormal[1.0, 0.15]	Usoskin NM inter-station spread ($\sim 15\%$)
LIS_norm	LogNormal[1.0, 0.12]	Voyager/AMS-02 calibration spread ($\sim 12\%$)
cross_sec	LogNormal[1.0, 0.20]	Nuclear σ tables: $\sim 20\%$
neutron_H	LogNormal[1.0, 0.25]	HZETRN neutron table: $\sim 25\%$ (Slaba et al., 2014)
hze_norm_scale	LogNormal[1.0, 0.20]	ACE/CRIS inter-instrument spread (George et al., 2009; Boschini et al., 2020)
Q_factor	LogNormal[1.0, $\ln 2$]	ICRP-60 $Q(L)$ form uncertainty: factor-of-2
DDREF	Uniform[1.0, 2.0]	BEIR VII / ICRP-103 range
ERR_scale	Normal[1.0, 0.30], clipped at 0	Cucinotta (2013) ERR fit residuals
EAR_scale	Normal[1.0, 0.35], clipped at 0	Cucinotta (2013) EAR fit residuals

Results compare to the NASA 30-day BFO limit of 250 mGy (Cucinotta and Durante, 2010). Mission encounter probability is Poisson, $P_{\text{SEP}}(T) = 1 - \exp(-\lambda T/365.25)$, $\lambda = 0.8$ events/yr at solar maximum, dropping 3–5 $\times$ at solar minimum (Feynman, Spitale, Wang and Gabriel, 1993).

1.8. Uncertainty Quantification via Latin Hypercube Sampling

Nine parameters are varied via LHS (Table 6): five physics (solar modulation, LIS normalization, nuclear cross-sections, neutron yield, HZE composition) and four biology (quality factor, DDREF, ERR scale, EAR scale).

Lognormal parameters cannot drive their model input negative; ERR_scale and EAR_scale are clipped at zero (affects $< 0.1\%$ of draws, no visible effect on percentiles). For each LHS sample, a full mission dose integration is performed (spectrum, transport, organ dose, quality factor, REID). Publication results use $N = 500$ (~ 90 min, single core, unparallelized); automated tests use $N = 20$ (UNCERTAINTY_SAMPLES env var).

Sensitivity is estimated via normalized Spearman ρ^2 , $S_i = \rho_i^2 / \sum_j \rho_j^2$. This is an approximation to first-order Sobol indices that underestimates interaction effects but is tractable at modest sample counts. A proper Sobol analysis would need $\sim 10^4$ samples (Saltelli, Annoni, Azzini, Campolongo, Ratto and Tarantola, 2010).

The value-of-information sweep re-runs the $N = 500$ ensemble once per parameter, halving that parameter's spread (sigma, or DDREF's range width centered on the same mean) while holding all others at baseline (narrow argument to `gcrisk.uncertainty.lhs_samples, scripts/sensitivity_narrowing_sweep.py`), recording the resulting change in p95 REID.

1.9. Dose-Averaged LET and Proton RBE Self-Consistency Check

Dose-averaged LET per species: $\bar{L}_{D,s} = \int \Phi_s S_s^2 dE / \int \Phi_s S_s dE$; field value $\bar{L}_D = \sum_s f_s \bar{L}_{D,s}$, $f_s = D_s / \sum_{s'} D_{s'}$.

The repository also contains a proton relative biological effectiveness (RBE) self-consistency layer using this same $\bar{L}_D$ infrastructure: Wedenberg et al. (2013) (Wedenberg, Lind and Hardemark, 2013) and McNamara et al. (2015) (McNamara, Schuermann and Paganetti, 2015), matched to OpenTOPAS-RBE scorer conventions and checked against OpenTOPAS-RBE reference values for a 100 MeV proton beam in water (V79, $\alpha/\beta = 1.41$ Gy). This is an implementation self-consistency check, and the foundation for a planned separate proton-therapy crossover study.

2. Supplementary Discussion: Full Limitations

2.1. Transport Model Approximations

- **Nuclear fragmentation secondaries.** Heavy-ion nuclear interactions are treated as simple fluence removal. As such, the pipeline doesn't take into account the lighter-ion secondaries produced through nuclear fragmentation. This leads to the secondary proton/helium flux being underestimated at depth and overestimates iron attenuation, largest above 30 g cm^{-2} . Based on the general direction/magnitude of differences between simplified transport and full fragmentation codes (Wilson et al., 1991; Slaba et al., 2014), my order-of-magnitude estimate is that omitting secondary production overestimates Q_{eff} at 16 g cm^{-2} by roughly 15–25%. This is my own rough extrapolation from those sources' general direction and magnitude. The resulting REID overestimate is likely similar in fractional magnitude, well within the $12\times \text{p5-p95}$ band but not negligible relative to the median.
- **Delta-ray production.** Secondary electrons from ion tracks are not tracked; small correction for macroscopic dose, would matter for microscopic track-structure DNA-damage models.
- **Geometry simplification.** Spacecraft modeled as a uniform aluminum slab; real geometries need ray-tracing (OLTARIS, Geant4).
- **Stopping power uncertainties.** NIST PSTAR/ASTAR accurate to $\sim 1\text{--}2\%$ above 1 MeV/n , less certain at lower energies (shell corrections).

2.2. Spectrum and Modulation

- **Monthly-resolution modulation cannot resolve daily variability.** ϕ is monthly-resolution only, so the pipeline's prediction varies smoothly within a month. The raw daily Pearson r against real MSL/RAD telemetry is weak ($r = -0.127$, $n = 218$, not significant), but this figure is largely confounded by a documented SEP event (X5.4 flare, AR11429, 2012-03-07, one of the largest of solar cycle 24) that spikes RAD's raw dose rate $\sim 30\times$ above baseline for a few days. This pipeline models SEP separately from chronic GCR, so comparing the GCR-only model against telemetry containing a live SEP spike biases the daily-resolution test for those days. Spearman's ρ over the full series ($\rho = -0.007$) and Pearson r with the event window excluded ($r = -0.001$, $n = 209$) are both consistent with zero. `scripts/daily_phi_from_nm.py` tested whether giving the model genuine daily-resolution modulation (real Oulu neutron-monitor count rate, independently calibrated to ϕ on a 1995–2011 window that never touches the MSL cruise period) would rescue the correlation; it does not ($r = -0.127 \rightarrow -0.119$), which rules out monthly ϕ resolution and points to the SEP confound as the real explanation. The multi-month trend the model targets remains directionally and quantitatively consistent regardless. Daily-resolution claims should be read as reflecting the monthly trend only.
- **ISS cross-check tests modulation shape only.** The ISS orbits inside Earth's magnetosphere, attenuating GCR flux by an amount not modeled here (no geomagnetic cutoff/trapped-radiation transport). The multi-year cross-check is restricted to trend correlation and does not validate the pipeline for near-Earth or trapped-radiation environments.
- **Force-field approximation.** Ignores charge-sign-dependent drift/diffusion anisotropy; during solar minimum can modify the GCR spectrum below $\sim 500 \text{ MeV/n}$ by 10–20% (Potgieter, 2013). This is within the propagated LIS-normalization uncertainty but not explicitly modeled.
- **Missing species.** Only H, He, C, O, Si, Fe are modeled. Magnesium's contribution is estimated at $< 3\%$ of total dose (scaled from Boschini 2020 Mg LIS normalization); all unmodeled species (Mg, N, Ne, S, Ca, ...) combined are estimated at $< 5\%$.
- **Anomalous cosmic rays.** Not modeled; can contribute below $\sim 100 \text{ MeV/n}$ during solar minimum.

2.3. Biological and Risk Model

- **NASA 2023 risk framework.** This pipeline implements Cucinotta (2013); NASA's 2023 update introduces age/sex-dependent career limits and revised tissue weighting factors not yet implemented here.

Table 7

Normalized Spearman ρ^2 sensitivity indices ($N = 500$, 35-year-old male, 259-day transit, 16 g cm^{-2} Al). Values sum to 1 within each column by construction. Radiobiological parameters dominate REID variance; physics parameters dominate D and H variance.

Parameter	REID	D	H
phi_scale	0.080	0.793	0.080
LIS_norm	0.001	0.188	0.002
cross_sec	0.020	<0.001	0.008
neutron_H	0.009	0.001	0.007
hze_norm_scale	0.008	0.014	0.009
Q_factor	0.749	0.001	0.886
DDREF	0.051	0.002	0.003
ERR_scale	0.040	0.001	0.001
EAR_scale	0.043	<0.001	0.005

Table 8

Value-of-information sweep: reduction in p95 REID from halving one parameter's uncertainty at a time (baseline p95 = 20.0%).

Parameter	Narrowed p95	Δ p95	% of baseline p95
Q_factor	13.3%	−6.7 pts	−33.5%
DDREF	19.1%	−0.9 pts	−4.5%
phi_scale	19.5%	−0.5 pts	−2.7%
ERR_scale	19.8%	−0.3 pts	−1.3%
LIS_norm	19.8%	−0.2 pts	−1.1%
hze_norm_scale	19.9%	−0.2 pts	−0.8%
neutron_H	20.0%	−0.1 pts	−0.3%
EAR_scale	20.0%	−0.0 pts	−0.1%
cross_sec	20.0%	0.0 pts	0.0%

- **High-LET extrapolation uncertainty.** EAR/ERR were fit primarily to low-LET atomic-bomb-survivor data with high-LET scaling from rodent experiments; no direct human epidemiological data exist for the HZE component of GCR. DDREF, bridging low-dose-rate space exposure and the acute exposures in the underlying datasets, is among the most debated parameters in space radiation risk assessment (Durante and Cucinotta, 2011).
- **Non-cancer risks.** Only cancer mortality is modeled; central nervous system (CNS) effects, cataracts, cardiovascular risk, and heritable effects are not quantified (CNS effects have drawn attention from rodent studies showing cognitive impairment below the cancer threshold (Cucinotta, Alp, Sulzman and Wang, 2014)).
- **Sex and age scope.** Results shown for 35-year-old male/female; a 25-year-old female has $\sim 1.5\text{--}2\times$ higher REID than a 45-year-old male for the same exposure (more remaining life-years, high breast/ovarian tissue weighting). The pipeline supports arbitrary age/sex; broader demographic analysis is future work.

3. Supplementary Tables

The tool-comparison table (Ours vs. HZETRN, OLTARIS, Geant4, TOPAS) is reproduced as Table 2 in the main text (§Comparison with Existing Tools and Reproducibility) rather than duplicated here.

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